

A Model of International Political Economy with Four Nations, Five Reserve Currencies, and a Five-State Political Dynamic

Foundations, Axioms and Mathematical Architecture

Soumadeep Ghosh

Kolkata, India

Abstract

This paper develops a mathematical model of International Political Economy built upon three structural principles. First, the international system contains at most four informationally independent sovereign nations under a common recoverable-money architecture. Second, no more than five reserve currencies can simultaneously remain globally viable under finite reserve rents and minimum military wage requirements. Third, every nation evolves according to a five-state political process consisting of Peace, Experiment, War, Stagnation and Absorb. The resulting framework combines economics, finance, monetary economics, political science, international relations, psychology, welfare economics, warfare economics, information theory, computer science, artificial intelligence, stochastic processes, network theory and dynamic optimization into a unified mathematical architecture. This paper develops the axiomatic foundations of the model together with its state space, mathematical structure, and network representation.

The paper ends with “The End”

Contents

1	Introduction	1
2	Literature	2
2.1	Economics	2
2.2	Finance	2
2.3	Political Science	2
2.4	International Relations	2
2.5	Computer Science	3
3	Objectives	3
4	Axioms	3
5	Primitive Objects	4
6	The Four Nations	4
7	Five Reserve Currencies	4
8	Political States	5
9	State Space	5
10	Network Architecture	5
11	Discussion	5
12	Future Development	6
13	Conclusion	6

1 Introduction

Modern International Political Economy traditionally studies interactions among

- sovereign governments,
- international financial markets,
- central banks,
- military institutions,
- multinational corporations,
- households,
- international organizations.

Although each discipline has developed sophisticated individual theories, a unified mathematical framework remains comparatively rare. Economic models generally assume political stability. Political science frequently abstracts from monetary structure. International relations often omit endogenous reserve-currency dynamics. Military strategy rarely incorporates monetary equilibrium. Artificial intelligence typically models agents independently of sovereign monetary systems. The present work seeks to integrate these perspectives into one coherent mathematical framework. The model is motivated by three structural observations.

1. International monetary systems appear capable of supporting only a finite number of globally dominant reserve currencies.
2. Political systems exhibit persistent transitions between peace, innovation, conflict and decline.
3. Information processing capacity imposes limits upon the number of informationally independent sovereign actors that can simultaneously participate within a common institutional architecture.

The resulting model represents the world economy as a coupled stochastic system consisting of

- four sovereign nations,
- five reserve currencies,
- five political states,
- endogenous monetary interactions,
- endogenous geopolitical interactions,
- dynamic feedback between economics and warfare.

Rather than treating economics and geopolitics as independent disciplines, they become two projections of a common dynamical system.

2 Literature

The present framework draws ideas from numerous disciplines.

2.1 Economics

General equilibrium theory studies allocation of scarce resources under decentralized markets. Dynamic macroeconomic models describe intertemporal allocation. International macroeconomics studies exchange rates, capital flows and monetary policy. The present work extends these ideas by introducing endogenous political-state transitions.

2.2 Finance

Reserve currencies, sovereign debt markets and financial intermediation provide the monetary backbone of the international system. Rather than treating reserve currency status as exogenous, the model allows geopolitical outcomes to influence reserve demand.

2.3 Political Science

Political science studies

- institutions,
- constitutions,
- governments,
- elections,
- state formation.

The present framework introduces explicit mathematical state transitions.

2.4 International Relations

Traditional theories include

- realism,
- liberal institutionalism,
- constructivism,
- balance-of-power theory.

The present model incorporates strategic interaction through stochastic state transitions.

2.5 Computer Science

Finite state machines and Markov chains naturally describe political evolution. Modern artificial intelligence similarly models environments as state-transition systems.

3 Objectives

The primary objectives are

1. construct a mathematically rigorous international political economy,
2. integrate monetary economics with geopolitical dynamics,
3. construct a finite-state political model,
4. unify reserve currencies and sovereign behaviour,
5. provide a foundation suitable for statistical estimation,
6. provide a foundation suitable for reinforcement learning,
7. provide a framework for future empirical testing.

4 Axioms

The model begins with the following axioms.

Axiom 1

There exist at most four informationally independent sovereign nations participating within a common recoverable monetary architecture.

Axiom 2

There exist at most five globally viable reserve currencies.

Axiom 3

Every sovereign nation occupies exactly one political state at every instant.

Axiom 4

Political states evolve according to stochastic transition probabilities.

Axiom 5

Military capability influences reserve-currency demand.

Axiom 6

Reserve-currency demand influences military capability through seigniorage.

Axiom 7

Economic production depends upon political state.

Axiom 8

International interactions generate endogenous transitions between political states.

5 Primitive Objects

The primitive mathematical objects are

Symbol	Meaning
N	Set of nations
C	Set of reserve currencies
S	Political state space
T	Trade network
A	Alliance network
R	Reserve allocations
M	Military capability
Y	Economic production

6 The Four Nations

Let

$$N = \{N_1, N_2, N_3, N_4\}.$$

Each nation possesses

- government,
- military,
- economy,
- financial system,
- political state.

The collection

$$N = \{N_i\}_{i=1}^4$$

constitutes the complete sovereign structure of the model.

7 Five Reserve Currencies

Let

$$C = \{C_1, C_2, C_3, C_4, C_5\}.$$

Each reserve currency possesses

- issuer,
- reserve demand,
- international credibility,
- military support,
- financial infrastructure.

Every nation may hold every reserve currency.

8 Political States

Define

$$\mathcal{S} = \{P, E, W, S, A\}.$$

The states are

State	Interpretation
P	Peace
E	Experiment
W	War
S	Stagnation
A	Absorb

Each nation occupies exactly one state.

9 State Space

The complete state of nation i is

$$X_i = (S_i, Y_i, M_i, R_i, A_i, T_i).$$

The world state becomes

$$X = (X_1, X_2, X_3, X_4).$$

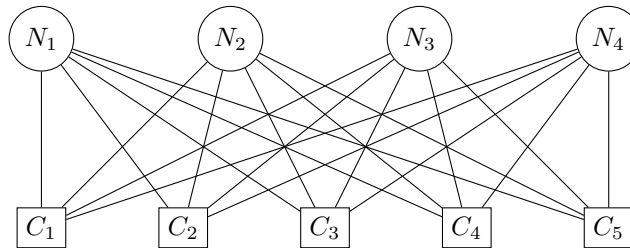
The political state space therefore contains

$$5^4 = 625$$

possible political configurations.

10 Network Architecture

The system naturally forms a multilayer network.



The upper layer represents sovereign nations. The lower layer represents reserve currencies. Edges represent reserve holdings.

11 Discussion

The resulting architecture possesses several desirable properties.

1. The number of nations is finite.
2. The number of reserve currencies is finite.
3. Political dynamics are discrete.
4. Economic variables evolve continuously.
5. The framework naturally combines graph theory and stochastic processes.
6. The architecture is immediately suitable for simulation.

12 Future Development

Subsequent parts of this manuscript shall develop

- stochastic transition matrices,
- dynamic programming,
- welfare economics,
- reserve currency equilibrium,
- warfare economics,
- geopolitical equilibria,
- reinforcement learning,
- Markov games,
- statistical estimation,
- empirical implementation.

13 Conclusion

Part I has established the mathematical foundations of a new model of International Political Economy. Rather than beginning from isolated markets or isolated governments, the framework begins from a coupled system consisting of four sovereign nations, five reserve currencies, and five political states. This finite architecture provides a tractable mathematical foundation upon which increasingly sophisticated dynamic models can be constructed in subsequent parts.

References

- [1] Kenneth J. Arrow. *The Limits of Organization*.
- [2] Gerard Debreu. *Theory of Value*.
- [3] Paul A. Samuelson. *Foundations of Economic Analysis*.
- [4] Robert E. Lucas. *Studies in Business Cycle Theory*.
- [5] Robert C. Merton. *Continuous-Time Finance*.
- [6] Claude E. Shannon. *A Mathematical Theory of Communication*.
- [7] Martin L. Puterman. *Markov Decision Processes*.
- [8] John G. Kemeny and J. Laurie Snell. *Finite Markov Chains*.

- [9] Kenneth Waltz. *Theory of International Politics*.
- [10] Robert Gilpin. *War and Change in World Politics*.
- [11] Charles Kindleberger. *The World in Depression*.
- [12] Barry Eichengreen. *Exorbitant Privilege*.

Glossary

Absorb Terminal political state.

Alliance Network Graph describing geopolitical cooperation.

Experiment Political state representing institutional innovation.

Finite State Machine Discrete stochastic system governing political evolution.

International Political Economy Joint study of economics and geopolitical interaction.

Peace Stable political state.

Reserve Currency Internationally accepted monetary asset.

Stagnation State of low economic and institutional growth.

War State involving organized military conflict.

The End